



Chapter 2

Market Economics

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Market, demand, supply and relationship

A **market** is a place (physical or online) where buyers and sellers meet to exchange goods or services.

It does not have to be a physical place.

It can be:

- Local market
- Financial market (like Nepal Stock Exchange)
- Online market (like Amazon)

Market: A physical or digital space where buyers and sellers interact to exchange goods and services.

Demand: The quantity of a product that consumers are willing and able to purchase at various price points.

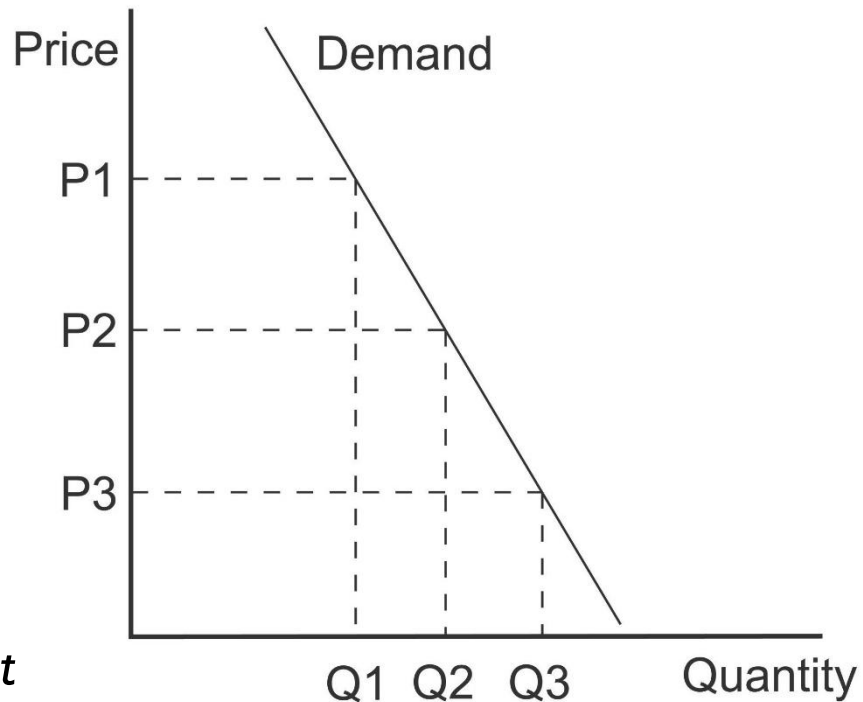
Supply: The total amount of a specific good or service that producers are willing and able to provide to the market at given prices.

Law of Demand

Law of Demand states that, other things remaining constant, quantity demanded of a good decreases when its price increases, and increases when its price decreases.

Price	Demand
↑ Increases	↓ Decreases
↓ Decreases	↑ Increases

Demand curve slopes downward because buyers buy less at higher prices.

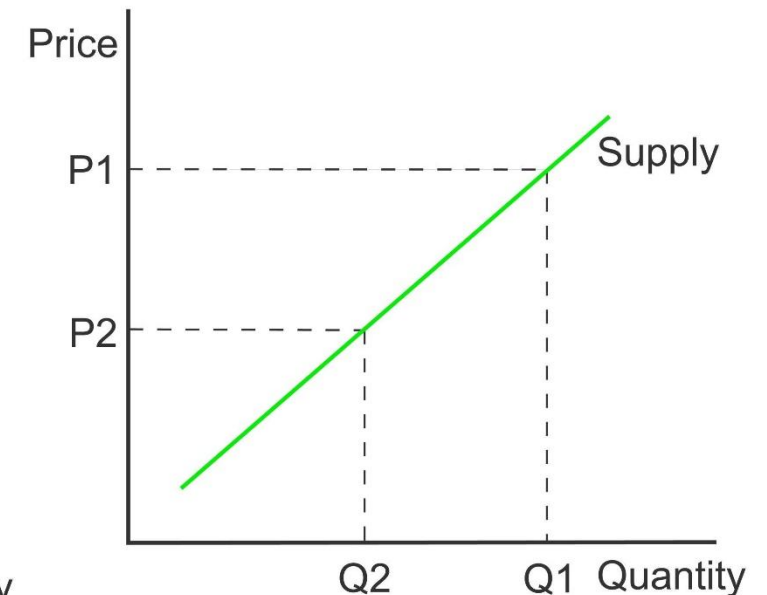
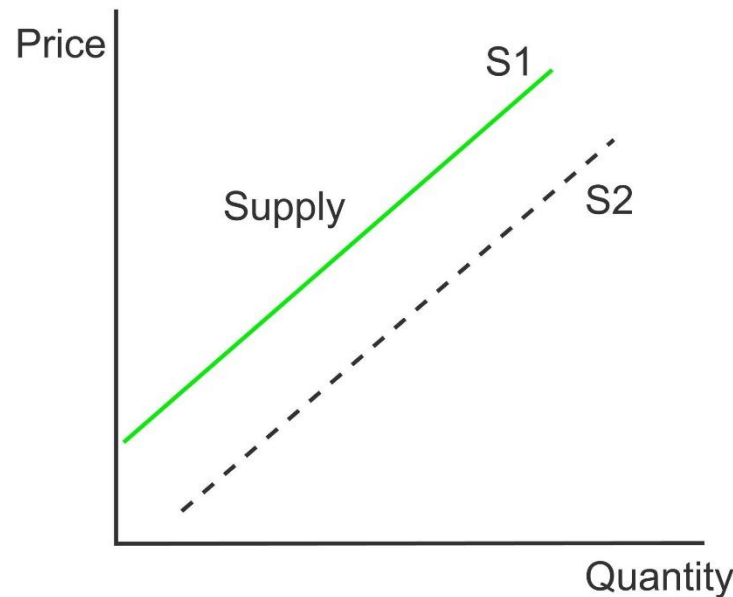


Law of Supply

Law of Supply states that, other things remaining constant, quantity supplied of a good increases when its price increases, and decreases when its price decreases.

Price	Supply
↑ Increases	↑ Increases
↓ Decreases	↓ Decreases

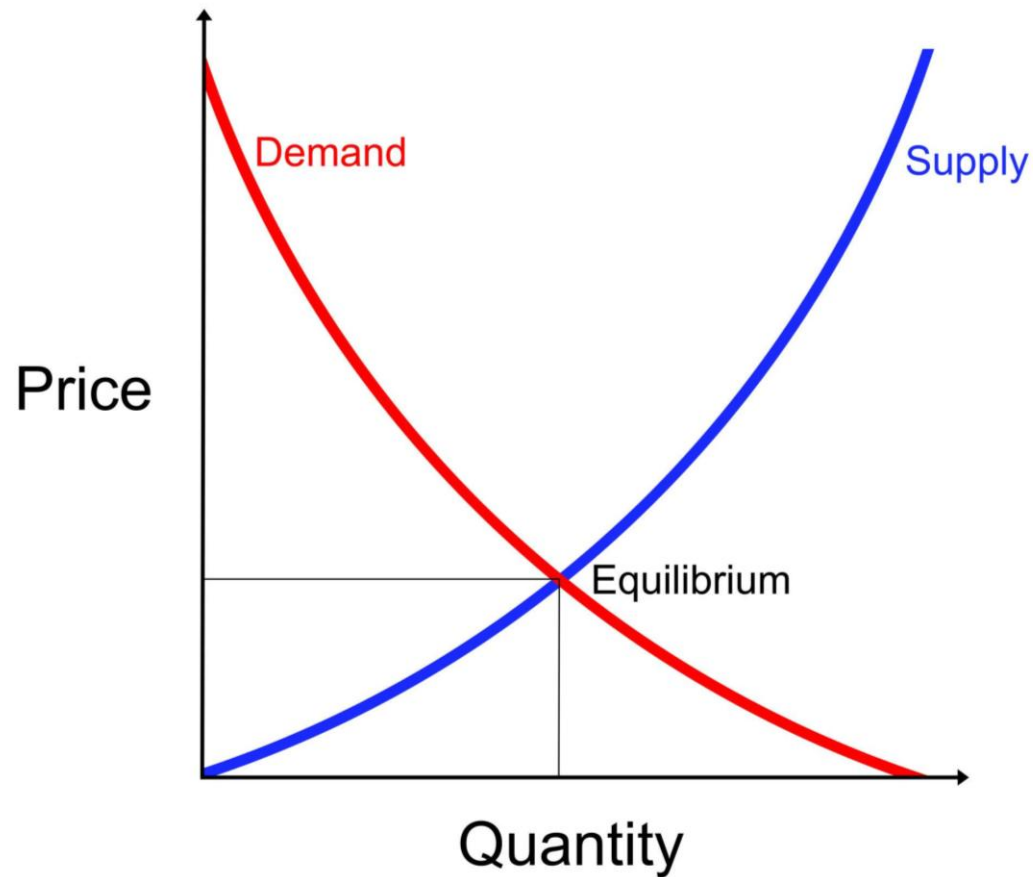
- Higher price results More profit
- Producers are encouraged to produce more
- Firms expand production



Supply curve slopes upward because sellers sell more at higher prices.

Basis of Comparison	Demand	Supply
Definition	Quantity of a good that consumers are willing and able to buy at a given price during a given period.	Quantity of a good that producers are willing and able to sell at a given price during a given period.
Concerned With	Buyers (Consumers)	Sellers (Producers)
Relationship with Price	Inverse relationship	Direct relationship
Law	Price $\uparrow \rightarrow$ Demand $\downarrow$ Price $\downarrow \rightarrow$ Demand $\uparrow$	Price $\uparrow \rightarrow$ Supply $\uparrow$ Price $\downarrow \rightarrow$ Supply $\downarrow$
Curve Shape	Downward sloping	Upward sloping
Determining Factors	Income, taste, price of related goods, expectations	Cost of production, technology, number of firms, expectations
Objective	To maximize satisfaction (utility)	To maximize profit
Effect of Increase in Price	Buyers purchase less	Producers supply more
Effect of Decrease in Price	Buyers purchase more	Producers supply less
Role in Market	Represents demand side of market	Represents supply side of market

Market Equilibrium



Market Equilibrium

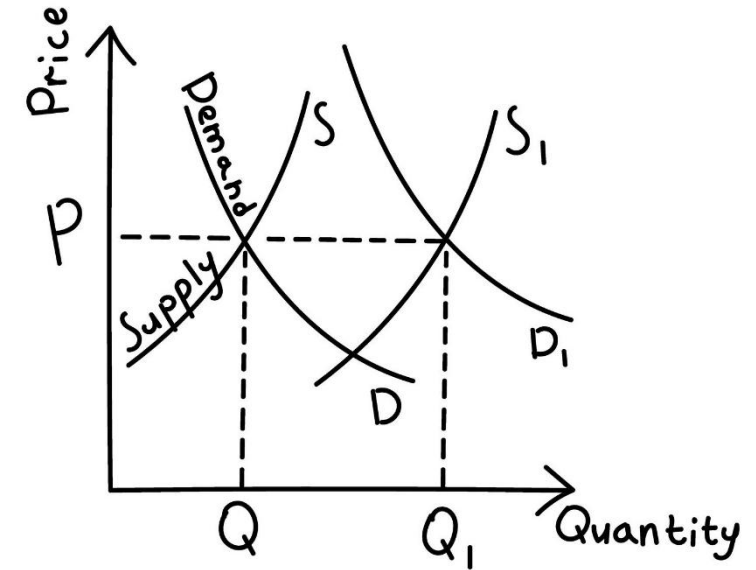
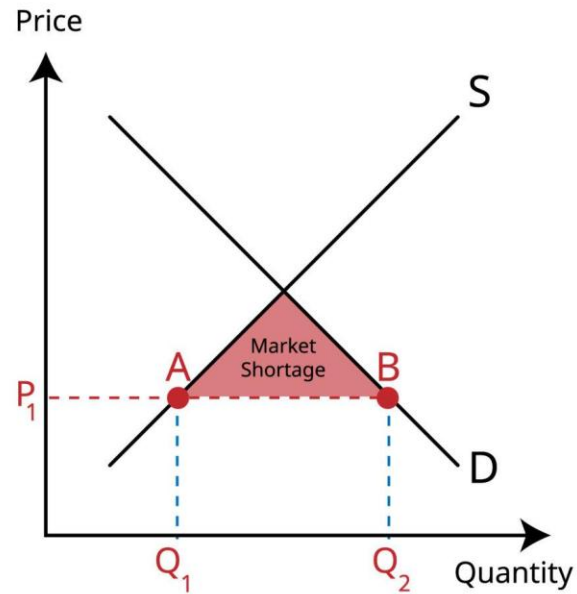
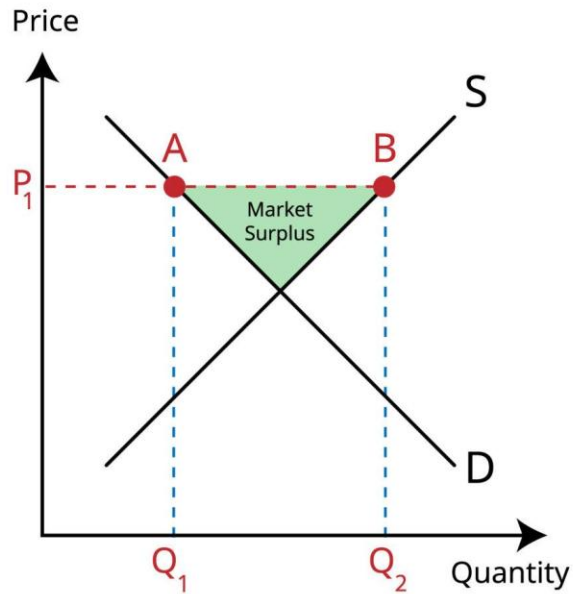
When Demand = Supply, the market is balanced.

- Equilibrium Price
- No shortage
- No surplus
- Stable market
- This price is called *Equilibrium Price (P_e)*

This quantity is called
Equilibrium Quantity (Q_e)

Market Disequilibrium

Market Disequilibrium

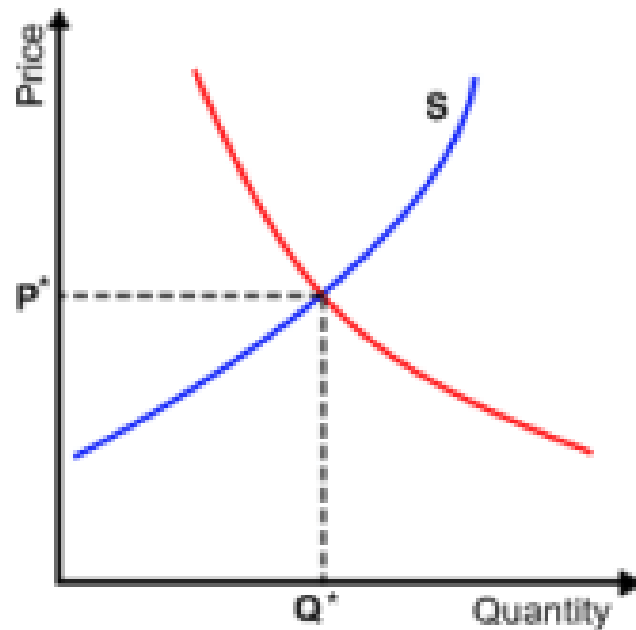


Relationship

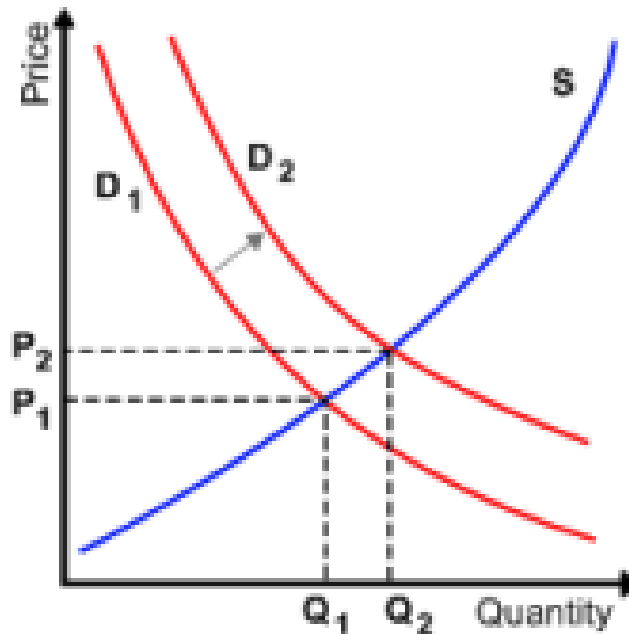
The relationship between supply and demand is the core of engineering economic analysis, guiding decision-making for projects and operations.

- *Feasibility Studies:* A primary application is in market feasibility analysis for new projects. Engineers determine if a proposed project is likely to succeed by ensuring a need for the product exists (demand) and the firm can produce it at a competitive cost (supply).
- *Price and Quantity Determination:* The interaction of supply and demand helps engineers and business managers determine the optimal equilibrium price and production quantity that balances customer willingness to pay with the firm's need for profit, avoiding shortages or surpluses.
- *Optimization:* Understanding these dynamics allows engineers to use tools like demand planning and price optimization models to maximize profits, minimize costs, and improve overall operational efficiency in complex supply chains. This includes using data analytics to understand how factors like price changes, promotions, and substitution effects influence market behavior.

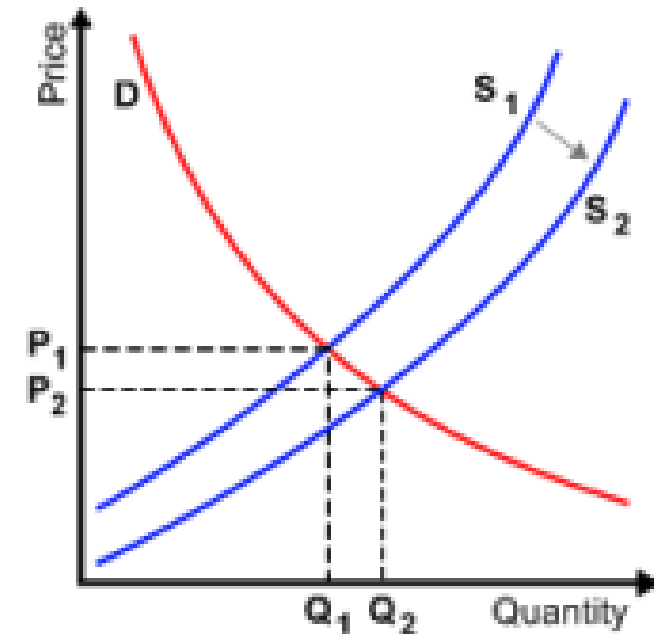
Relationship



(a) Price equilibrium at price P^* and quantity Q^* .



(b) An increase in demand causes an increase in price.



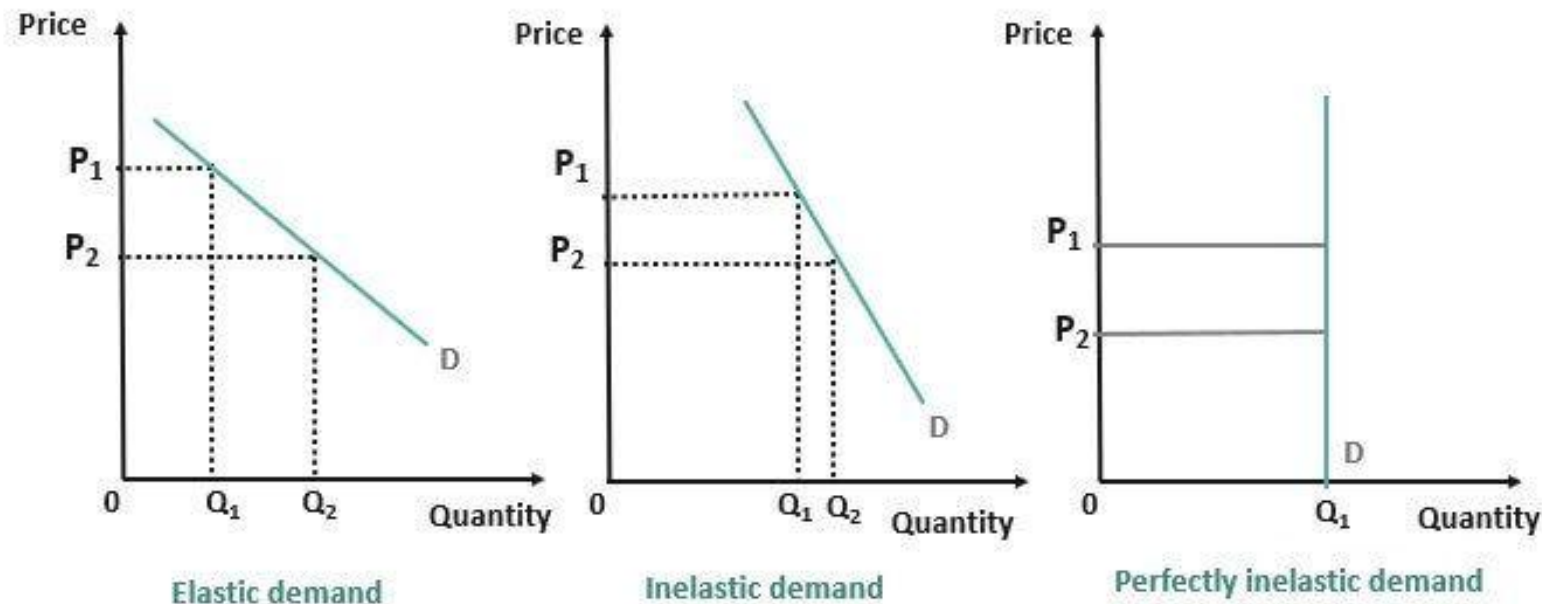
(c) An increase in supply causes a reduction in price.

Elasticity, application of elasticity and government policies

Elasticity measures the responsiveness of quantity demanded or supplied due to change in price or other factors.

Elasticity tells us how much demand or supply changes when price changes.

Elastic vs. Inelastic vs. Perfectly Inelastic Demand



Elasticity, application of elasticity and government policies

In engineering economics, elasticity measures the responsiveness of one economic variable to changes in another, typically focusing on how quantity demanded or supplied reacts to shifts in price, income, or other market conditions.

Types of Elasticity

Price Elasticity of Demand (PED): Measures how sensitive consumers are to changes in a product's price.

Elastic ($PED > 1$): Large changes in demand for small price shifts (e.g., luxury cars).

Perfect inelastic ($PED=0$): Demand remains stable despite price fluctuations (e.g., essential medicines).

Price Elasticity of Supply (PES): Measures how quickly producers can adjust output when prices change. Supply is more elastic in the long run as firms can expand capacity.

Income Elasticity of Demand (YED): Measures demand shifts relative to changes in consumer income, identifying goods as "normal" (demand rises with income) or "inferior" (demand falls as income rises).

Cross-Price Elasticity (XED): Assesses how the price change of one good affects the demand for another, identifying substitutes (positive XED) or complements (negative XED).

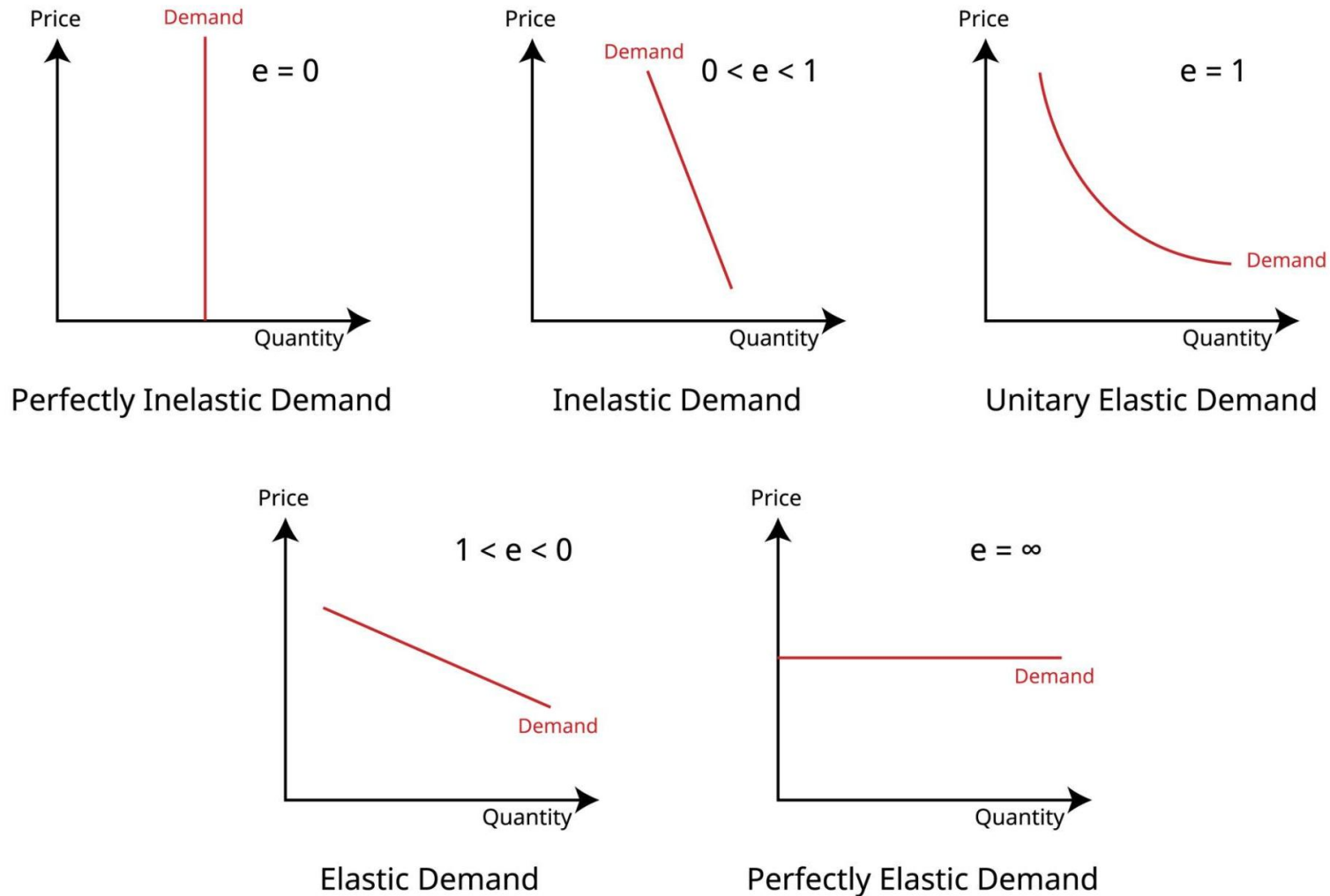
Elasticity, application of elasticity and government policies

$$E_d = \frac{\% \text{ Change in Quantity Demanded}}{\% \text{ Change in Price}}$$

Type	Condition	
Elastic	$E_d > 1$	Demand changes more than price
Inelastic	$E_d < 1$	Demand changes less than price
Unitary	$E_d = 1$	Equal change
Perfectly Elastic	$E_d = \infty$	Very sensitive (horizontal line)
Perfectly Inelastic	$E_d = 0$	No change (vertical line)

Elasticity, application of elasticity and government policies

Price Elasticity of Demand (PED)



Elasticity, application of elasticity and government policies

Type	Condition (E_d)	Curve Shape	Meaning	Example
Elastic Demand	$E_d > 1$	Flatter	Small price change → Large demand change	Luxury tiles, decorative materials
Inelastic Demand	$E_d < 1$	Steep	Large price change → Small demand change	Cement, steel
Unitary Elastic	$E_d = 1$	Normal slope	% change in price = % change in demand	Some household goods
Perfectly Elastic	$E_d = \infty$	Horizontal line	Even small price rise → Demand becomes zero	Perfect competition (theoretical case)
Perfectly Inelastic	$E_d = 0$	Vertical line	Quantity does not change at all	Life-saving medicines

Questions

A consumer demands 40 kg of a commodity when its price is Rs. 1 per kg.
If the price increases by Rs. 0.10, what would be the quantity demanded?

PED = -1

Original Quantity (Q) = 40

Original Price (P) = 1

New Quantity (Q_1) = ?

New Price (P_1) = 1.1

Change in Quantity (ΔQ) = ?

Change in Price (ΔP) = 0.1

Questions

Price Elasticity of Demand

$$PED = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

or

$$-1 = \frac{\Delta Q}{0.1} \times \frac{1}{40}$$

$$\Delta Q = -4$$

As price is increasing, the quantity demanded must decrease by 4.

$$\text{New Quantity} = \text{Initial Quantity} + \Delta Q$$

$$= 40 + (-4)$$

$$= 36$$

Elasticity, application of elasticity and government policies

Income Elasticity of Demand (YED)

Measures responsiveness of demand due to change in income.

$$E_y = \frac{\% \text{ Change in Quantity Demanded}}{\% \text{ Change in Income}}$$

Type	Meaning
Positive (>0)	Normal goods (demand↑ when income ↑)
Negative (<0)	Inferior goods (demand↓ when income ↑)

Elasticity, application of elasticity and government policies

Price Elasticity of Supply (PES)

Measures responsiveness of quantity supplied due to change in price.

$$E_s = \frac{\% \text{ Change in Quantity Supplied}}{\% \text{ Change in Price}}$$

Types under PES

- Elastic supply
- Inelastic supply
- Unitary supply
- Perfectly elastic supply
- Perfectly inelastic supply

Formulas

- **Equilibrium:** $Q_d = Q_s$
- **Price Elasticity (PED):** $\frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$
- **Market Demand:** $\sum (\text{Individual Demands})$
- **Income Elasticity (E_y):** $\frac{\% \Delta \text{Quantity}}{\% \Delta \text{Income}}$

Elasticity, application of elasticity and government policies

Consider the demand for a good. At price Rs 4, the demand for the good is 25 units. Suppose price of the good increases to Rs 5, and as a result, the demand for the good falls to 20 units. Calculate the price elasticity?

Original Quantity (Q) = 25 units Original Price (P) = ₹ 4

Fall in Quantity (ΔQ) = – 5 units Rise in Price (ΔP) = ₹ 1

New Quantity (Q_1) = 20 units New Price (P_1) = ₹ 5

Elasticity of Demand (ED) = ?

Price Elasticity of Demand (ED)

$$ED = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = \frac{-5}{1} \times \frac{4}{25} = -0.8$$

Demand is less elastic as $ED < 1$.

Negative sign indicates inverse relationship between price and quantity demanded.

Elasticity, application of elasticity and government policies

Suppose the price elasticity of demand for a good is -0.2. If there is a 5% increase in the price of the good, by what percentage will the demand for the good go down?

% Change in Demand = ? % Change in Price = 5%

Elasticity of Demand (ED) = (-)0.2

Price Elasticity of Demand (ED)

$$ED = \frac{\% \text{Change in Quantity Demanded}}{\% \text{Change in Price}}$$

$$-0.2 = \frac{\% \text{Change in Quantity Demanded}}{5\%}$$

$$\% \text{Change in Quantity Demanded} = -1\%$$

Percentage fall in demand = 1%

Demand for the good will fall by 1%.



For every 1% change in price, demand changes by 0.2% in opposite direction.

Elasticity, application of elasticity and government policies

If demand function for a commodity is given as $Q_d = 40 - 0.1P$ and supply function is given as $Q_s = (-20 + 0.2P)$, find the equilibrium price and quantity.

At equilibrium, $Q_d = Q_s$ Thus,

$$40 - 0.1P = -20 + 0.2P$$

$$40 + 20 = 0.2P + 0.1P$$

$$60 = 0.3P$$

$$P = \text{Rs } 200$$

$$\text{At } P = 200, Q_d = Q_s = 40 - 0.1(200) = 20 \text{ units}$$

Questions

ED is perfectly inelastic as quantity demanded does not change at all in response to change in price. Thus, its demand curve will be vertical / parallel to the y-axis. A consumer spends Rs. 1000 on a good priced at Rs. 10 per unit. When its price falls by 20 percent, the consumer spends Rs. 800 on the good. Calculate the price elasticity of demand by the Percentage method.

Initial Price (P) = 10

Initial Expenditure = 1000

Initial Quantity (Q) = 100

New Price (P_1) = 8

New Expenditure = 800

New Quantity (Q_1)

$$Q_1 = \frac{Exp}{Price} = \frac{800}{8} = 100$$

$$| \Delta P = (-)2 | \Delta Q = 0 |$$

Questions

Price Elasticity of Demand

$$PED = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$PED = \frac{0}{(-2)} \times \frac{10}{100}$$

$$PED = 0$$

ED is perfectly inelastic as quantity demanded does not change at all in response to change in price. Thus, its demand curve will be vertical / parallel to the y-axis.

Elasticity, application of elasticity and government policies

If demand function for a commodity is given as $Q_d = 600 - 2P$ and supply function is given as $Q_s = 3P$, make a schedule of demand and supply at prices Rs 80, Rs 100, Rs 120, Rs 140 and Rs 160. Also find the equilibrium price and quantity.

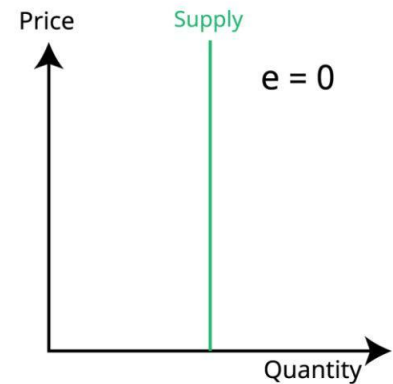
ANS:

Price	Quantity demanded	Quantity supplied
80	$600 - 2(80) = 440$	$3(80) = 240$
100	$600 - 2(100) = 400$	$3(100) = 300$
120	360	360
140	320	420
160	280	480

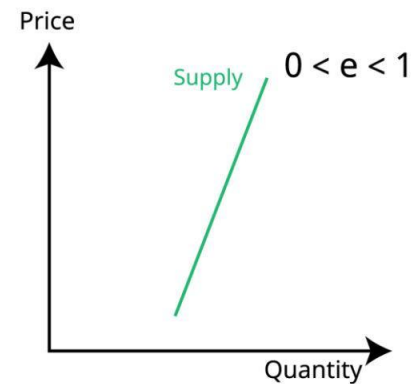
Equilibrium price is Rs 120. Equilibrium quantity is 360 units.

Elasticity, application of elasticity and government policies

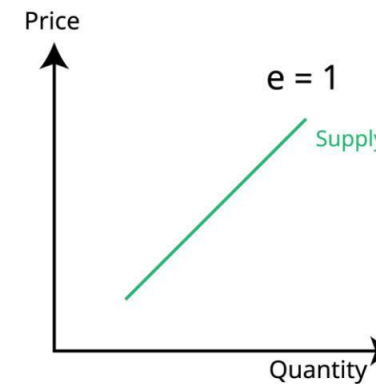
Price Elasticity of Supply (PES)



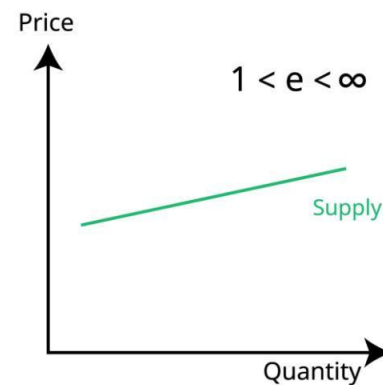
Perfectly Inelastic Supply



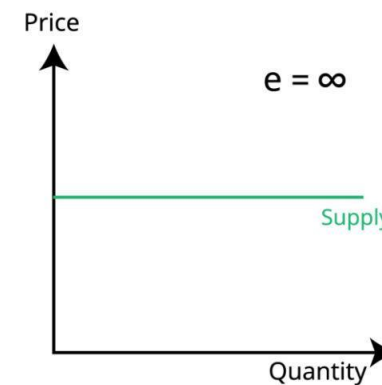
Inelastic Supply



Unitary Elastic Supply



Elastic Supply



Perfectly Elastic Supply

Elasticity, application of elasticity and government policies

Type	Condition (E_s)	Curve Shape	Meaning	Example
Elastic Supply	$E_s > 1$	Flatter upward	Small price rise → Large supply increase	Brick production
Inelastic Supply	$E_s < 1$	Steep upward	Price rise → Small supply increase	Steel plants (short run)
Unitary Supply	$E_s = 1$	Moderate slope	% change equal	Normal manufacturing
Perfectly Elastic Supply	$E_s = \infty$	Horizontal line	Unlimited supply at one price	Theoretical case
Perfectly Inelastic Supply	$E_s = 0$	Vertical line	Supply fixed	Land

Market Structure

Market Structure	Sellers	Price Control	Entry Barrier	Example
Perfect Competition	Many	None	No	Sand, bricks
Monopoly	One	High	Very High	Electricity
Monopolistic Competition	Many	Moderate	Low	Construction firms
Oligopoly	Few	High	High	Cement, steel
Duopoly	Two	High	High	Telecom
Monopsony	Many sellers, one buyer	Buyer controls	High	Govt projects

Elasticity, application of elasticity and government policies

Applications in Engineering Economics

- *Pricing Strategy*: Engineers and managers use elasticity to find the "equilibrium price" that maximizes revenue without causing sharp declines in sales volume.
- *Revenue Projections*: Elasticity models make revenue forecasts more realistic by accounting for buyer sensitivity.
- *Production Planning*: Understanding PES helps engineers decide when to invest in extra capacity or adjust inventory levels to avoid stockouts.
- *Risk Management*: Analyzing input price elasticity (e.g., raw materials) allows for better assessment of cost risks and exposure to market volatility.
- *Structural Material Selection*: In a physical engineering sense, elasticity refers to material properties like Young's Modulus, used to design structures (like bridges or crane ropes) that return to their original shape after being stressed.

Elasticity, application of elasticity and government policies

Relationship with Government Policies

- Governments utilize elasticity to design interventions that minimize market distortions while achieving social goals.
- Taxation: Lawmakers typically target inelastic goods (e.g., tobacco, gasoline) for higher taxes because consumers will keep buying them, ensuring stable tax revenue. Conversely, taxing elastic goods (e.g., sugary drinks) is often used to discourage consumption for public health.
- Subsidies: Policies like tax credits for electric vehicles are often applied to elastic goods to significantly boost adoption by making them more affordable.
- Price Controls:
- Price Ceilings: Setting a maximum price (e.g., rent control) often leads to shortages if the ceiling is below market equilibrium.
- Price Floors: Setting a minimum price (e.g., minimum wage) can lead to surpluses if set above equilibrium.
- Tax Incidence: The burden of a tax falls more heavily on the side of the market (buyers or sellers) that is more inelastic.

Externality and market inefficiency

An **externality** is a cost or benefit that affects a third party not directly involved in an economic transaction.

Externalities lead to market inefficiency because the market price does not reflect the true social costs or benefits, resulting in either overproduction or underproduction of goods and services.

Externalities create a divergence between private costs/benefits (borne by the producer/consumer) and social costs/benefits (borne by society as a whole). This causes market failure and inefficient allocation of resources.

Negative Externality	Harm to others	Pollution from factories
Positive Externality	Benefit to others	Road construction increases nearby property value

Externality and market inefficiency

Negative Externalities: These impose costs on third parties.

Examples: A factory polluting the air or water, a driver causing traffic congestion, or a loud party disturbing neighbors.

Inefficiency: Because producers/consumers don't bear the full social cost, they produce/consume more than the socially optimal amount, leading to overproduction.

Outcome: The market equilibrium quantity is too high, resulting in a misallocation of resources and a loss of social welfare.

Positive Externalities: These generate benefits for third parties.

Examples: A person getting vaccinated (reducing the spread of disease), a firm's R&D leading to new technologies benefiting society, or education creating a more skilled workforce.

Inefficiency: Because producers/consumers don't capture all the social benefits, they produce/consume less than the socially optimal amount, leading to underproduction.

Outcome: The market equilibrium quantity is too low, meaning society misses out on potential welfare gains

Externality and market inefficiency

Market inefficiency occurs when resources are not allocated in a way that maximizes total social welfare. The market does not produce the “right quantity” for society.

In an efficient market:

$$\textit{Marginal Social Benefit} = \textit{Marginal Social Cost}$$

At this point:

- Resources are properly used
- No waste
- No shortage
- Maximum social welfare

Where:

- MPC = Marginal Private Cost
- MSC = Marginal Social Cost
- MSB = Marginal Social Benefit

If this condition is not satisfied → Market inefficiency occurs.

Questions

A factory produces goods with the following cost and benefit functions:

$$MPC = Q$$

$$MSC = 2Q$$

$$MSB = 100 - Q$$

Questions

1. Free Market Equilibrium

(Private market outcome)

Free market equilibrium occurs when:

$$MPC = MSB$$

Substitute the functions:

$$Q = 100 - Q$$

$$2Q = 100$$

$$Q_m = 50$$

Questions

2. Socially Optimal Output

(Efficient outcome for society)

Social optimum occurs when:

$$MSC = MSB$$

Substitute the functions:

$$2Q = 100 - Q$$

$$3Q = 100$$

$$Q_s = 33.33$$

Questions

3. Interpretation

Outcome	Quantity
Free Market Output Q_m	50
Socially Optimal Output Q_s	33.33

The free market produces too much output:

$$50 > 33.33$$

This happens due to a **negative externality**, where private producers ignore external costs imposed on society.

Market failure and firm behavior

Market failure occurs when the free market fails to allocate resources efficiently, leading to social welfare loss.

Market does not give the best outcome for society.

Firm behavior is a primary driver of these failures, as companies acting in their own rational self-interest often create outcomes that are sub-optimal for society.

Firm behavior explains how firms make decisions regarding:

- Price
- Output
- Profit maximization Firms decide output where:
 $MR=MC$ (Marginal Revenue = Marginal Cost)

Market failure and firm behavior

Types of Market Failure

Market failures are typically categorized by the specific "distortion" preventing equilibrium:

- **Complete Market Failure:** The market fails to exist entirely, and no goods are supplied despite demand.
- **Partial Market Failure:** The market exists but produces the wrong quantity or at the wrong price.
- **Negative Externalities:** Social costs (private cost + external cost) exceed private costs, leading to over-consumption (e.g., smoking, pollution).
- **Positive Externalities:** Social benefits exceed private benefits, leading to under-consumption (e.g., vaccinations, education).

Questions

1. Demand and supply are given by:

$$Q_d = 100 - 2P, \quad Q_s = 20 + 3P$$

Find equilibrium price and quantity.

If the government imposes a tax of Rs 5 per unit on sellers, find the new equilibrium price and quantity.

Questions

Before tax

At equilibrium,

$$Q_d = Q_s$$

$$100 - 2P = 20 + 3P$$

$$80 = 5P$$

$$P = 16$$

Now quantity:

$$Q = 100 - 2(16) = 68$$

Answer before tax

- *Equilibrium price = Rs 16*
- *Equilibrium quantity = 68 units*

Questions

After tax

Let consumer price = P_c

Seller receives:

$$P_s = P_c - 5$$

Supply becomes:

$$Q_s = 20 + 3(P_c - 5)$$

$$Q_s = 20 + 3P_c - 15 = 5 + 3P_c$$

Demand:

$$Q_d = 100 - 2P_c$$

Questions

At equilibrium:

$$100 - 2P_c = 5 + 3P_c$$

$$95 = 5P_c$$

$$P_c = 19$$

Seller price:

$$P_s = 19 - 5 = 14$$

Quantity:

$$Q = 100 - 2(19) = 62$$

Answer after tax

- Consumer price = Rs 19
- Producer price = Rs 14
- New equilibrium quantity = 62 units

Questions

4. For a commodity: Find shortage in the market.

$$Q_d = 200 - 5P, \quad Q_s = 50 + 5P$$

Government sets a **price ceiling** at Rs 12.

Demand

$$Q_d = 200 - 5(12) = 200 - 60 = 140$$

Supply

$$Q_s = 50 + 5(12) = 50 + 60 = 110$$

Shortage

$$\text{Shortage} = Q_d - Q_s = 140 - 110 = 30$$

Questions

1. A product's price increases by 10% and quantity demanded decreases by 15%. Calculate price elasticity of demand.
2. Income of consumers increases from Rs 40,000 to Rs 50,000 per month, and demand for a commodity rises from 100 units to 130 units. Find the income elasticity of demand.
3. The supply of a good increases from 200 units to 260 units when price rises from Rs 10 to Rs 12. Calculate the price elasticity of supply.
4. At price Rs 50, quantity demanded is 100 units. When price falls to Rs 48, quantity demanded rises to 110 units. Find price elasticity of demand.

Questions

5. The demand for a good increases by 20% due to a fall in its price.
If the price elasticity of demand is -0.8 , calculate the percentage fall in price.
6. The quantity demanded of a commodity is 500 units when the price is Rs. 8 per unit.
When the price falls to Rs. 6 per unit, the quantity demanded increases to 600 units.
Calculate the price elasticity of demand.
7. When the price of a good is Rs. 12 per unit, the consumer buys 24 units. When the price increases to Rs. 14 per unit, the consumer buys 20 units.
Calculate the price elasticity of demand.

Questions

8. A commodity has:

old income = Rs 25,000

new income = Rs 30,000

old demand = 40 units

new demand = 52 units

Find income elasticity.

9. A firm observes that a 5% reduction in price causes a 20% increase in demand. Calculate price elasticity.

Questions

Consider a competitive market for which the quantities demanded and supplied (per year) at various prices are given as follows:

Price (Rs.)	Demand (millions)	Supply (millions)
60	22	14
80	20	16
100	18	18
120	16	20

- Calculate the price elasticity of demand when the price is Rs.80, and Rs.100.
- Calculate the price elasticity of supply when the price is Rs. 80, and Rs.100.
- What are the equilibrium price and quantity?
- Suppose the government sets a price ceiling of Rs. 80. Will there be a shortage, and, if so, how large will it be?

Questions

If the demand and supply curve for computers is: $D = 100 - 6P$, $S = 28 + 3P$. Where P is the price of computers, what is the quantity of computers bought and sold at equilibrium?

Initially, the supply curve for a product is represented by $S(p) = 60 + 4p$. The demand curve is represented by $D(p) = 150 - 2p$. If the supply curve shifts to $S(p) = 80 + 4p$, calculate the new equilibrium price and quantity.

Tom's supply equation for selling handmade mugs is as follows:

$$Q = 5 + 1.5P$$

How many mugs will he sell if the price is Rs. 2 a mug? What if the price is Rs. 4 a mug?

The quantity demanded of Good Z depends upon the price of Z (P_z), monthly income (Y), and the price of a related Good W (P_w). Demand for Good Z (Q_z) is given by equation 1: $Q_z = 150 - 8P_z + 2Y - 15P_w$. Find the demand equation for Good Z in terms of the price for Z (P_z), when Y is \$50 and $P_w = \$6$.

Consider the following demand and supply functions for a good.

$$\text{Quantity demanded} = 160 - 2p$$

$$\text{Quantity supplied} = -40 + 2p$$

Questions

Given the demand function $Q(p) = 400 - 2p$, find the quantity demanded when the price is \$60.

For two goods, A and B, the demand functions are:

$$Q_A = 200 - 3p$$

$$Q_B = 150 - 2p$$

Determine the cross elasticity of demand when $p_A = 30$ and 40 and $p_B = 40$ and 50 . Interpret the result.

Initially, the supply curve for a product is represented by $S(p) = 60 + 4p$. The demand curve is represented by $D(p) = 150 - 2p$. If the supply curve shifts to $S(p) = 80 + 4p$, calculate the new equilibrium price and quantity.

Consider a demand function $Q(p) = 1000 / (p^2)$, where p is the price. Find the quantity demanded when the price is \$10.

Thank you for your attention!